higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

WORKSHOP PRACTICE
NQF LEVEL 2

(12041012)

8 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 It is important that employees are made aware of the dangers in their workplace by training them.

List FIVE topics that training should include. (5)

1.2 Two common causes of accidents and incidents in the work environment are personal factors and work factors.

Name THREE work factors. (3)

1.3 Chemical hazards, biological hazards, ergonomic hazards and physical hazards are all occupational-health hazards.

Name ONE example of each of the given hazards. (4 × 1) (4)

1.4 The aim of the Occupational Health and Safety Act of 1993 is to provide for the health and safety of employees at work.

Name TWO other parties excluded by the OHSA. (2)

1.5 Give the full name of the grading system NOSA uses as a framework for the development and management of factories and workshops. (2)

1.6 Define scaffolding. (1)

1.7 List THREE safety rules to follow when working on an elevated position. (3)

[20]

QUESTION 2

2.1 State whether the following are TRUE or FALSE. Write only correct answer next to the number 2.1.1 – 2.1.6 in the ANSWER BOOK.

2.1.1 In order to assess whether the casualty is breathing, the first aider should keep the casualty's head in the tilted position.

2.1.2 Workers should not be trained in first- aid.

2.1.3 Always wear gloves when attending to casualty.

2.1.4 In order to prevent the inhalation of possible vomit, the casualty's mouth must be cleared out.

2.1.5 Use a mouthpiece during artificial respiration (AR) to promote the transmission of diseases.

2.1.6 Shock is a condition in which blood circulation is disturbed

(6 × 1) (6)

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2.2 Before commencing any form of treatment, a first aider should first monitor the casualty's vital signs to establish his/her condition by following the C-A-B-C scheme.

What does the acronym C-A-B-C stand for? (4 × ½) (2)

2.3 Define resuscitation. (1)

2.4 Some of the most common and serious types of accidents are burns and scalds.

2.4.1 Name THREE types of burns (3)

2.4.2 Give an example for each one you have chosen. (3)
[15]

QUESTION 3

3.1 Match the following files in COLUMN A with their applications in COLUMN B. Write only correct letter (A–E) next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

COLUMN A	COLUMN B
3.1.1 Half- Round	A Enlarge square and rectangular holes
3.1.2 Square	B Widen holes
3.1.3 Round	C General surface filing
3.1.4 Flat	D Enlarge triangular holes in a work piece
3.1.5 Triangular	E Finish internal round corners

(5)

3.2 Pliers are made in different shapes and sizes to perform specific operations.

Give FOUR uses of pliers when working with thin metal or wire. (4)

3.3 A hand tap is a tool used to cut an internal thread into a drilled hole.

Name the THREE groups of taps. (3)

3.4 As an electrician you will be required to use a bending spring at some time.

List THREE inspections to observe before using a bending spring. (3)
[15]

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QUESTION 4

4.1 Study the power tool shown in FIGURE 2 and answer the questions.

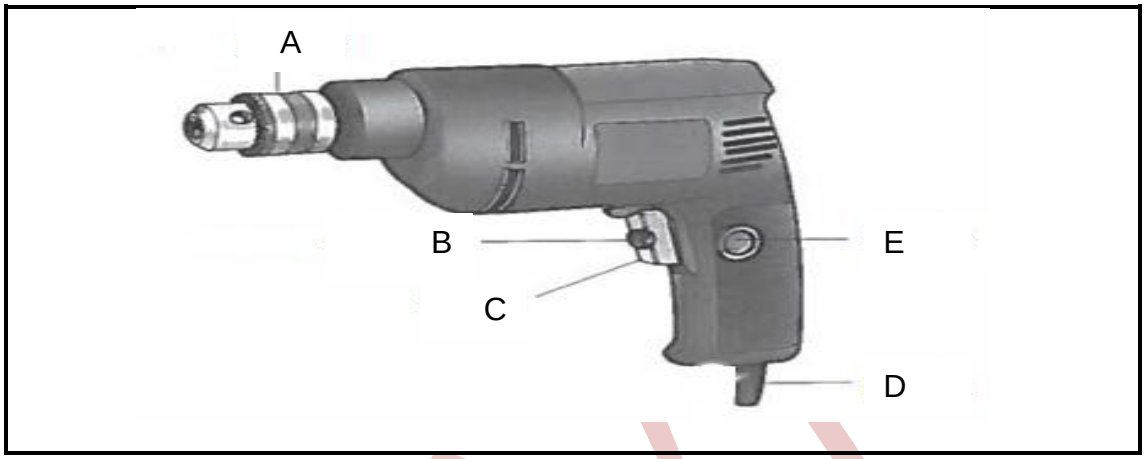


FIGURE 2

4.1.1 What is the name of the power tool in FIGURE 2? (1)

4.1.2 Label the parts of the power tool in FIGURE 2 by writing only the answer next to the letter (A–E) in the ANSWER BOOK. (5 × 1) (5)

4.1.3 What type of drill bit would you use to drill through concrete using power tool in Question 4.1.1.? (2)

4.2 It is important to inspect all power tools before using them to make sure they are in good working condition.

List FOUR safety checks when inspecting an angle grinder before using it. (4)

4.3 All power tools must be maintained, cared for and stored properly according to the manufacturer's instructions to ensure they operate correctly and last longer.

4.3.1 Give TWO ways to take care of an orbital sander. (2)

4.3.2 How would you store an orbital sander to keep it in good working condition? (1)

[15]

QUESTION 5

5.1 Different types of measuring instruments are used in the engineering field for various applications.

Name FOUR types of measuring instruments, excluding a vernier calliper. (4)

5.2 5.2.1 Before using measuring equipment it is important to ensure their accuracy by inspecting them first.

List FIVE factors that affect the accuracy of a vernier calliper. (5)

5.2.2 Name TWO steps how to store a vernier calliper to keep it in good working condition. (2)

5.3 Sketch and label the main parts of the spirit level. (4)

[15]

QUESTION 6

6.1 Briefly describe the function of the Labour Relations Act. (2)

6.2 A standard SABS symbolic safety sign consists of a geometric shape, a colour and a picture (pictogram).

What is the purpose of symbolic safety signs? (3)

6.3 Overhead cranes are essential equipment used in most types of industry.

6.3.1 What are overhead cranes used for? (1)

6.3.2 Name TWO types of shackles (2)

6.4 Planning and organising are very important aspects of any electrical task. Without planning and organising one would not know where to start or continue.

What is the result of poor planning or no planning? (2)

[10]

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QUESTION 7

7.1 Complete the sentence by filling in the missing words. Write your answer next to the number 7.1(a) –7.1(d) in the ANSWER BOOK.

When wires or cables are (a)… together, they should have the same (b) … conductivity and (c)…strength they had before they were (d)… (4)

7.2 Define the term *de-soldering*. (2)

7.3 Before using a new soldering-iron tip, it should be tinned.

7.3.1 Why do we have to tin the new soldering-iron tip? (1)

7.3.2 Name THREE steps on how to tin a new soldering-iron tip. (3)

[10]

TOTAL: 100

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